Md. Jubaer Khan

Senior Software Engineer — Embedded Systems & C/C++

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PROFESSIONAL SUMMARY

Senior Software Engineer with over five years of experience delivering embedded, desktop and web software, including four-plus years at BJIT Ltd. building safety-critical systems for Japanese and international clients. Strong in C and C++ across RTOS and Embedded Linux targets, automotive and industrial communication protocols (CAN 2.0B, SAE J1939), and MISRA-C / MISRA-C++ / BARR-C compliant development. Experienced in requirement analysis, system architecture, code review and mentoring within Agile/Scrum teams. Recent work spans AI-agent-based tooling and legacy modernisation from VB6 to C#/.NET. Recognised as BJIT Best Performer of the Year for 2023–2024, and co-author of a peer-reviewed IEEE Transactions paper on deep-learning-based COVID-19 detection.

TECHNICAL SKILLS

Languages	C, C++ (Embedded C/C++, Visual C++), Python, C#, VB.NET, Java (core), JavaScript
Embedded & RTOS	RTOS, Embedded Linux (Yocto, Buildroot), STM32 ecosystem, ESP32/ESP8266, embedded system design, peripheral and sensor interfacing, schematic and datasheet analysis
Protocols	CAN 2.0B, SAE J1939, UART, SPI, I2C, HDLC, RFID, Wi-Fi (IEEE 802.11)
Standards & Practice	MISRA-C, MISRA-C++, BARR-C, design patterns, UML, OOAD, test-driven development, code and design review
Desktop & Web	Qt Framework, MFC, .NET Framework / .NET Core, ASP.NET MVC, Razor, Flask, Django, HTML, CSS, Bootstrap
Databases	MySQL, MS SQL Server
Tools & Process	Git, GitHub, Bitbucket, Jira, Confluence, Visual Studio, VS Code, PlatformIO, Arduino IDE, CodeWarrior, Orchestra Suite, Agile/Scrum
AI & ML	AI agents and agentic workflows, machine learning and NLP fundamentals (TensorFlow, Keras, CNN, Word2vec)
Other	Unity, LaTeX, Figma

PROFESSIONAL EXPERIENCE

Senior Software Engineer *Jul 2025 – Present*

BJIT Ltd. • Dhaka, Bangladesh

- Provide technical leadership for embedded and PC application projects, owning system architecture and critical modules.
- Lead design and code reviews, enforcing MISRA-C, MISRA-C++ and BARR-C compliance across project teams.
- Mentor junior and mid-level engineers through design walkthroughs, technical guidance and best-practice sharing.
- Work with product owners and stakeholders on requirement analysis, effort estimation and technical risk mitigation.
- Resolve complex system-level defects in RTOS and Embedded Linux environments and support integration phases.

Software Engineer *Apr 2022 – Jun 2025*

BJIT Ltd. • Dhaka, Bangladesh

- Designed, implemented, tested and deployed embedded, desktop and web software components for client products.
- Developed communication-driven applications over CAN 2.0B, SAE J1939, UART, SPI and I2C.
- Built custom Embedded Linux distributions with Yocto and Buildroot for target hardware platforms.
- Delivered cross-platform desktop applications in C++/Qt and PC applications on .NET (C#, VB.NET).
- Implemented service-oriented embedded architectures and produced requirement and design documentation.
- Collaborated in Agile/Scrum teams to meet delivery deadlines and resolve design and coding issues.

Web Developer & Designer (part-time) *Nov 2021 – Feb 2022*

Independent client engagement

- Planned, built, deployed and maintained a client business website with WordPress and cPanel, including professional email setup.

Intern Software Engineer *Feb 2021 – Jun 2021*

North South University — Office of Information Technology

- Led a three-member team to build an NLP-based student assistance chatbot in Python and Flask, deployed live on university infrastructure.
- Led development of an RFID-based automated printing and access facility using C++, ESP32 and the RC522 reader over SPI.
- Developed prototypes and ran feasibility studies on emerging technologies for academic and administrative systems.

Student Worker, IT Support Team *Aug 2018 – Jan 2021*

North South University — Office of Information Technology

- Provided technical support to faculty, officials and students, and staffed the IT service desk and request tracking.
- Prepared and maintained PCs for printing and engineering labs, covering OS and software installation, IP and domain setup.
- Configured, troubleshot and maintained the PaperCut print management server and print driver estate.
- Provided technical support for university programming contests, classrooms and other IT services.

SELECTED PROFESSIONAL PROJECTS

Client and product names are withheld under confidentiality agreements.

AI Agent-Based Monitoring Software *Jul 2026 – Aug 2026*

In-house research and development of AI-agent-based monitoring software.

Tech: AI agents • In-house R&D

Agentic HDLC Communication *Jul 2026 – Aug 2026*

In-house research and development applying an agentic approach to HDLC link-layer communication.

Tech: HDLC, AI agents • In-house R&D

Legacy Management Software Modernisation — VB6 to C#/.NET *Nov 2025 – Present*

Migration of a legacy VB6 management application to a web-based C#/.NET system, including analysis of existing functionality and re-implementation.

Tech: VB6, C#, .NET, ASP.NET, web

Automotive Embedded Control Platform — 2nd generation *Feb 2025 – Nov 2025*

Design and development of safety-critical embedded modules for automotive-grade vehicle systems, from analysis and detailed design through production and test. Contributed to architecture discussions, code review and system integration.

Tech: C, C++, CAN 2.0B, SAE J1939, RTOS • Team of 5

Internal R&D, PC Application OJT & Engineering Support Platforms *Nov 2023 – Present*

Development of internal tools, proof-of-concepts and training platforms supporting on-the-job training and experimental work across embedded and desktop stacks.

Tech: C#, C/C++, RTOS, Embedded Linux, Unity • Team of 19

Custom Embedded Linux Distribution *Nov 2023 – Mar 2024*

Built and customised Embedded Linux images tailored to specific target hardware, including layer configuration, package selection and image tuning.

Tech: Yocto, Buildroot, Linux • Team of 2

Cross-Platform Desktop Application (Qt) *Sep 2023 – Oct 2023*

Developed a high-performance cross-platform GUI application, covering requirement analysis, detailed design and implementation.

Tech: C++, Qt Framework • Team of 2

PC & Handheld Terminal Application Suite *Jul 2023 – Aug 2023*

Built PC and handheld-terminal applications for an industrial data capture workflow, including UI and device-side logic.

Tech: VB.NET, RFgen Mobile Development Studio • Team of 3

Document Editor & Viewer Feature Development *May 2023 – Jun 2023*

Contributed PDF editing and viewing features to a large desktop product, working across driver, COM add-in and GDI rendering layers.

Tech: C#, C++, COM, GDI • Team of 21

Web-Controlled Four-Wheel Robot *May 2023 – Jun 2023*

Designed and implemented Wi-Fi remote control and a web interface for a four-wheel robotic platform.

Tech: NodeMCU ESP8266, Wi-Fi, C++ • Individual

Enterprise Asset Management Web Application *Mar 2023 – May 2023*

Implemented web-based asset management features with role-based access control, from investigation and basic design through detailed design.

Tech: ASP.NET MVC, C#, Razor, Bootstrap • Team of 6

Automotive Embedded Control Platform — 1st generation *Jun 2022 – Feb 2023*
Developed vehicle interface management software for an automotive client, covering analysis, detailed design, production and test under embedded coding standards.
Tech: C, C++, CAN 2.0B, Python • Team of 6

Automated Teller Machine Control System *May 2022 – Jun 2022*
Contributed to investigation, analysis and basic design of an ATM control system on an Embedded Linux target with I2C peripheral interfacing.
Tech: C++, Embedded Linux, I2C, UML • Team of 5

EDUCATION

B.Sc. in Computer Science & Engineering *May 2017 – May 2021*
North South University, Dhaka, Bangladesh
CGPA 3.59 / 4.00 • Specialisation coursework: Machine Learning, Pattern Recognition & Neural Networks, Theory of Computation, Mobile & Wireless Application Development

Higher Secondary Certificate (HSC), Science *2013 – 2015*
Jhalokathi Govt. College, Jhalokathi, Bangladesh

RESEARCH & PUBLICATIONS

XCR-Net: A Computer Aided Framework to Detect COVID-19 *2024*
Ashik Mostafa Alvi, **Md. Jubaer Khan**, Nishat Tasnim Manami, Zubair Azim Miazi, Kate Wang, Siuly Siuly, Hua Wang
IEEE Transactions on Consumer Electronics, vol. 70, no. 4, pp. 7551–7561 • doi.org/10.1109/TCE.2024.3446793

Building COVID Knowledge Base from Medical Journals *2021*
Md. Jubaer Khan, Walidul Alam Ehab
Undergraduate thesis, Dept. of Electrical & Computer Engineering, North South University • repository.northsouth.edu

AWARDS & ACHIEVEMENTS

Best Performer of the Year — BJIT Ltd. *2023–2024*
Company-wide annual recognition, presented at the BJIT General Meeting 2023–2024.

CERTIFICATIONS & TRAINING

Fundamental Engineering Course in Embedded System *Apr – Jul 2022*
BJIT Academy — three-month technical skill training programme.

Artificial Intelligence Fundamentals *Jun 2026*
IBM SkillsBuild — covering machine learning and deep learning, NLP and computer vision, generative AI, AI ethics, and IBM Watson Studio.

Information Security (IS-101) *Dec 2024*
BJIT Academy — information security training and assessment programme.

ACADEMIC & PERSONAL PROJECTS

Retinal Blood Vessel Segmentation using Deep Learning *Jan 2020 – Jun 2020*
Applied CNN architectures to segment retinal blood vessels for disease detection, with normalisation, CLAHE and gamma-adjustment pre-processing to improve accuracy. Responsible for background study, dataset collection, result analysis and reporting.
Tech: Python, TensorFlow, Keras, CNN • Team of 2

Lung Disease Detection from X-Ray Images *Sep 2020 – Dec 2020*
Led a five-member team building a CNN-based classifier for lung disease detection from chest X-ray images, covering investigation, analysis and basic design.
Tech: Python, CNN, image processing • Team lead, team of 5

MenuWhiz — NFC-Based Restaurant Ordering System *Apr 2020 – Aug 2020*
Led development of an NFC-driven ordering platform combining an Android customer app, a restaurant web portal and a shared menu/order database. Built the web front end, Android app and UML/ER design.
Tech: Android, NFC, Django, Python, HTML, CSS, JavaScript, REST, SQL • Team lead, team of 3

Line Follower Robot for Office Assistance *Jan 2021 – Feb 2021*
Led a three-member team building an autonomous line-following robot with IR and motion sensing and load balancing for in-office delivery tasks.

Tech: IR & motion sensors, autonomous driving, embedded C • Team lead, team of 3

Office Management System for IT Support *Jun 2017 – Aug 2017*

Built a C-based office management tool to organise officials, assign projects and track work progress, covering file handling, core functions and account type design.

Tech: C, file handling • Team of 3

LANGUAGES & ADDITIONAL INFORMATION

Training	Embedded Systems engineering course, BJIT Academy (2022) • AI Fundamentals, IBM SkillsBuild (2026) • Information Security IS-101, BJIT Academy (2024)
Languages	Bangla — native • English — fluent (reading, writing, speaking)
Personal	Date of birth 13 December 1998 • Nationality: Bangladeshi

References available on request.